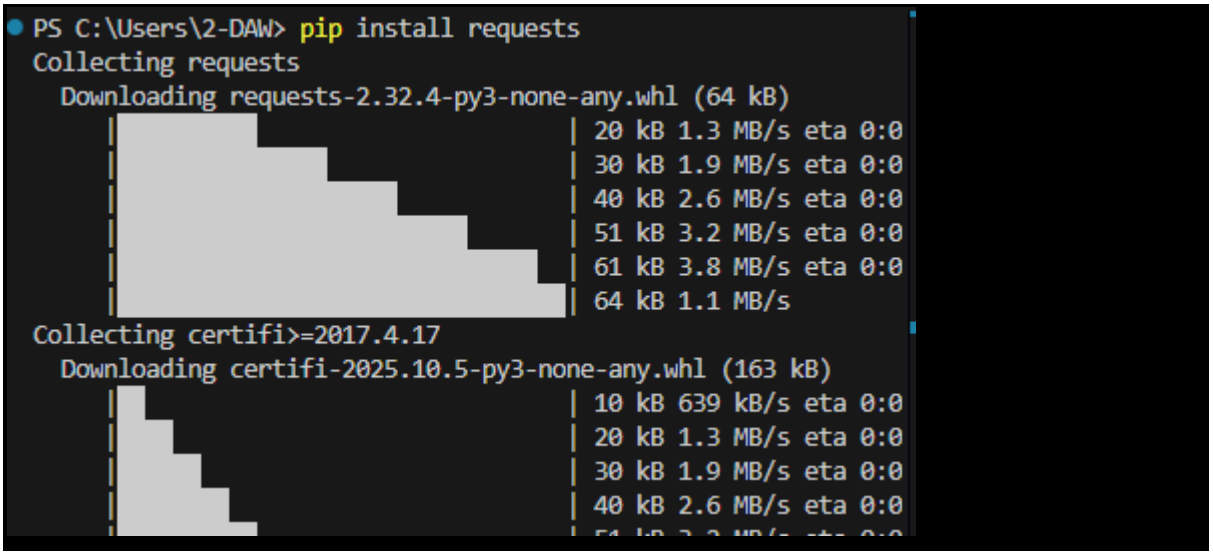
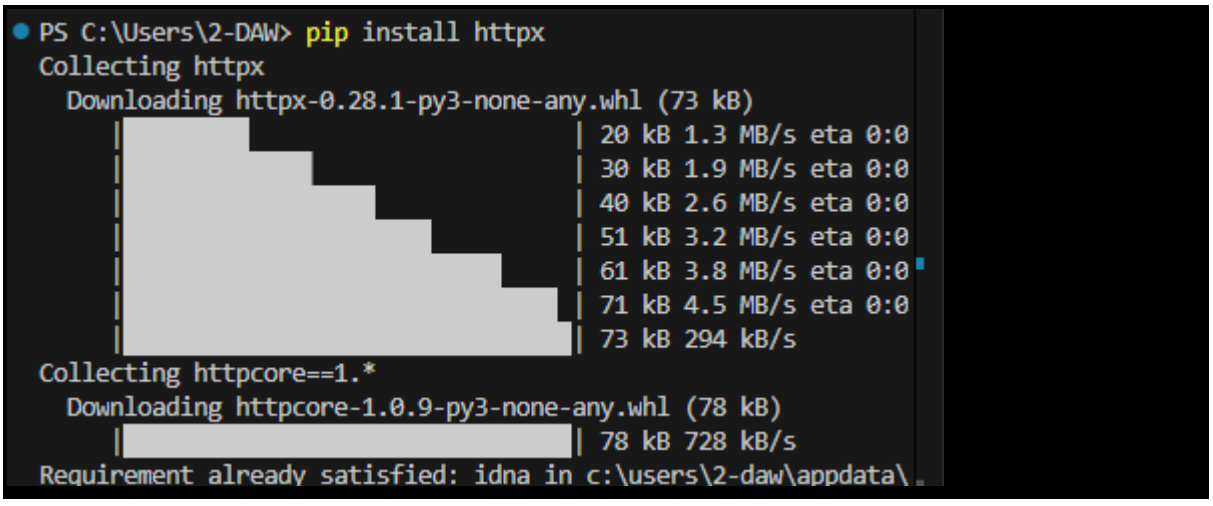
Web scraping / automatización web

- **requests**
  Sirve para **conectarte a páginas o APIs** por HTTP(s) y descargar su contenido (HTML, JSON, archivos). Es la base del scraping cuando la web **no necesita JavaScript** para mostrar los datos. Puedes enviar cabeceras, autenticación y formularios.

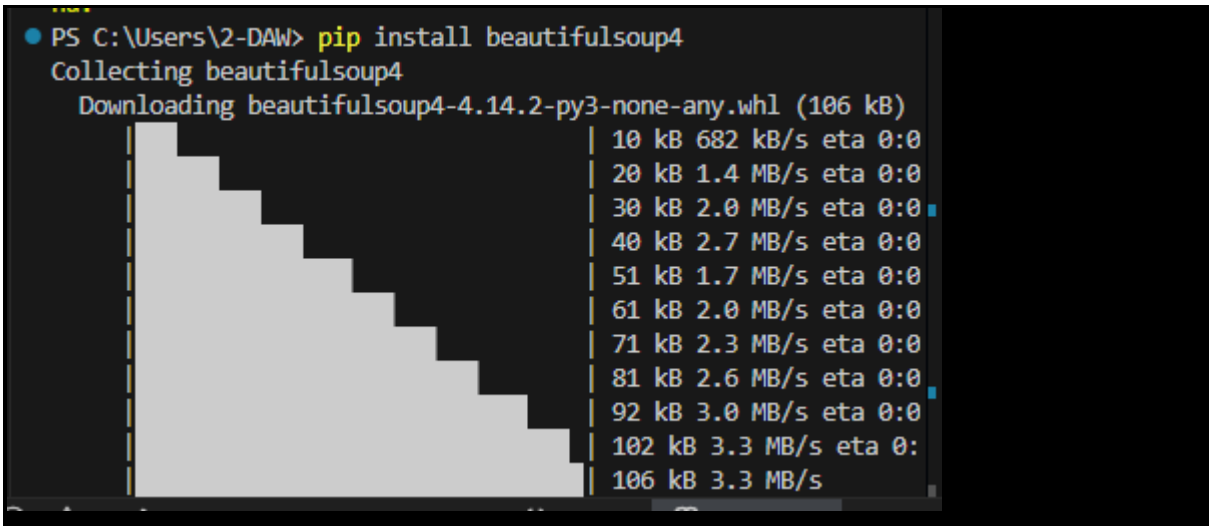
```
pip install requests
```



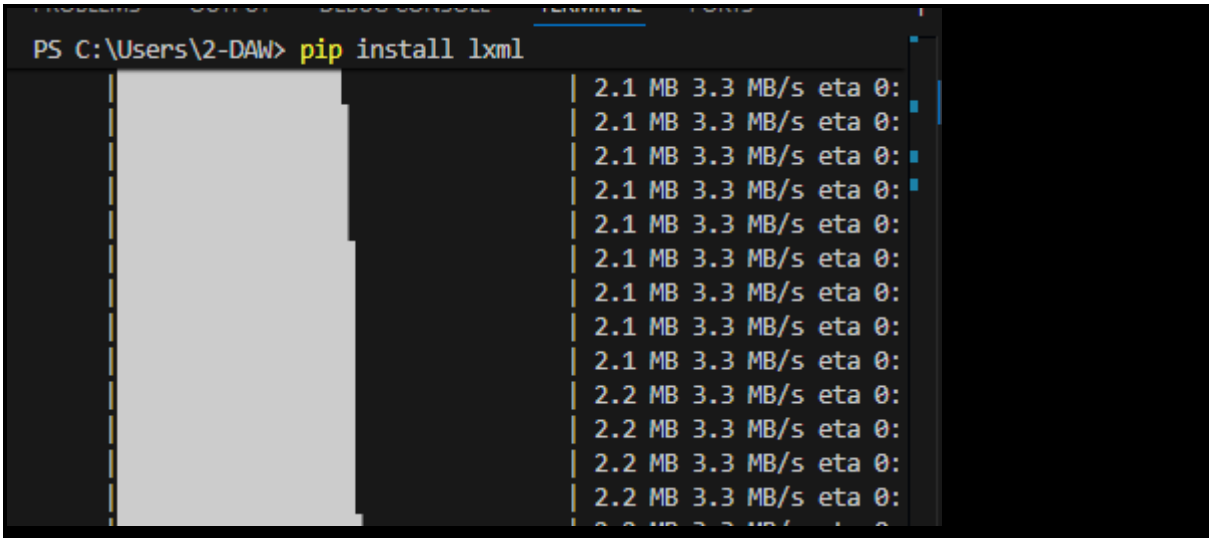

```
pip install httpx
```



pip install beautifulsoup4



pip install lxml



pip install parsel

```
PS C:\Users\2-DAW> pip install parsel
Requirement already satisfied: lxml in c:\users\2-daw\appdata\
local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2kfra8
p0\localcache\local-packages\python38\site-packages (from pars
el) (6.0.2)
Installing collected packages: w3lib, packaging, jmespath, css
select, parsel
Successfully installed cssselect-1.2.0 jmespath-1.0.1 packagin
g-25.0 parsel-1.9.1 w3lib-2.2.1
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install selectolax

```
PS C:\Users\2-DAW> pip install selectolax
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s
Installing collected packages: selectolax
Successfully installed selectolax-0.3.29
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
PS C:\Users\2-DAW> pip install selectolax
```

pip install scrapy

```
PS C:\Users\2-DAW> pip install scrapy
Requirement already satisfied: certifi>=2017.4.17 in c:\users\
2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3
.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packa
ges (from requests>=2.1.0->tldextract->scrapy) (2025.10.5)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\
users\2-daw\appdata\local\packages\pythonsoftwarefoundation.py
thon.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site
-packages (from requests>=2.1.0->tldextract->scrapy) (3.4.4)
Installing collected packages: pycparser, tomli, setuptools, p
yasn1, cffi, zope.interface, requests-file, pyasn1-modules, it
emadapter, incremental, hyperlink, filelock, cryptography, con
stantly, automat, attrs, Twisted, tldextract, service-identity
, queuelib, pyOpenSSL, PyDispatcher, protego, itemloaders, def
usedxml, scrapy
```

pip install playwright
playwright install

```
es (from pyee==12.0.0->playwright) (4.13.2)
Installing collected packages: pyee, greenlet, playwright
  WARNING: The script playwright.exe is installed in 'C:\Users
\2-DAW\AppData\Local\Packages\PythonSoftwareFoundation.Python.
3.8_qbz5n2kfra8p0\LocalCache\local-packages\Python38\Scripts'
which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to
suppress this warning, use --no-warn-script-location.
Successfully installed greenlet-3.1.1 playwright-1.48.0 pyee-1
2.0.0
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
```

```
PS C:\Users\2-DAW> playwright install
PS C:\Users\2-DAW> pip install playwright
Requirement already satisfied: playwright in c:\users\2-daw\ap
pdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n
2kfra8p0\localcache\local-packages\python38\site-packages (1.4
8.0)
Requirement already satisfied: pyee==12.0.0 in c:\users\2-daw\
appdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz
5n2kfra8p0\localcache\local-packages\python38\site-packages (f
rom playwright) (12.0.0)
Requirement already satisfied: greenlet==3.1.1 in c:\users\2-d
aw\appdata\local\packages\pythonsoftwarefoundation.python.3.8_
qbz5n2kfra8p0\localcache\local-packages\python38\site-packages
 (from playwright) (3.1.1)
Requirement already satisfied: typing-extensions in c:\users\2
```

pip install selenium

```
  WARNING: The script wsdump.exe is installed in 'C:\Users\2
AW\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.
qbz5n2kfra8p0\LocalCache\local-packages\Python38\Scripts' wh
h is not on PATH.
  Consider adding this directory to PATH or, if you prefer t
suppress this warning, use --no-warn-script-location.
Successfully installed outcome-1.3.0.post0 pysocks-1.7.1 sel
ium-4.27.1 sortedcontainers-2.4.0 trio-0.27.0 trio-websocket
.12.2 websocket-client-1.8.0 wsproto-1.2.0
WARNING: You are using pip version 21.1.1; however, version
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppDat
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' com
```

pip install cloudscraper

```
PS C:\Users\2-DAW> pip install cloudscraper
  Collecting cloudscraper
    Downloading cloudscraper-1.2.71-py2.py3-none-any.whl (99 kB)
       |▌                               | 20 kB 1.4 MB/s eta 0:0
       |▉                               | 30 kB 2.0 MB/s eta 0:0
       |▏                               | 40 kB 2.7 MB/s eta 0:0
       |▋                               | 51 kB 3.4 MB/s eta 0:0
       |▊                               | 61 kB 4.1 MB/s eta 0:0
       |▍                               | 71 kB 2.3 MB/s eta 0:0
       |▌                               | 81 kB 2.6 MB/s eta 0:0
       |▏                               | 92 kB 3.0 MB/s eta 0:0
       |▎                               | 99 kB 291 kB/s
  Collecting pyparsing>=2.4.7
    Downloading pyparsing-3.1.4-py3-none-any.whl (104 kB)
```

pip install fake-useragent

```
PS C:\Users\2-DAW> pip install fake-useragent
  Downloading importlib_resources-6.4.5-py3-none-any.whl (36 k
B)
Collecting zipp>=3.1.0
  Downloading zipp-3.20.2-py3-none-any.whl (9.2 kB)
Installing collected packages: zipp, importlib-resources, fake
-useragent
Successfully installed fake-useragent-2.0.0 importlib-resource
s-6.4.5 zipp-3.20.2
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install trafilatura

```
PS C:\Users\2-DAW> pip install trafilatura
  Downloading htmldate-1.9.3-py3-none-any.whl (31 kB)
Requirement already satisfied: urllib3<3,>=1.26 in c:\users\2-
daw\appdata\local\packages\pythonsoftwarefoundation.python.3.8
_qbz5n2kfra8p0\localcache\local-packages\python38\site-package
s (from trafilatura) (2.2.3)
Requirement already satisfied: lxml>=5.3.0 in c:\users\2-daw\a
ppdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5
n2kfra8p0\localcache\local-packages\python38\site-packages (fr
om trafilatura) (6.0.2)
Collecting tld>=0.13
  Downloading tld-0.13.1-py2.py3-none-any.whl (274 kB)
     |                                | 274 kB ...
Collecting babel>=2.16.0
  Downloading babel-2.17.0-py3-none-any.whl (10.2 MB)
     |                                | 10.2 MB ...
```

pip install newspaper3k

```
PS C:\Users\2-DAW> pip install newspaper3k
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.3 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 6.8 MB/s eta 0:
     |                                | 7.4 MB 100 kB/s
```

pip install pandas

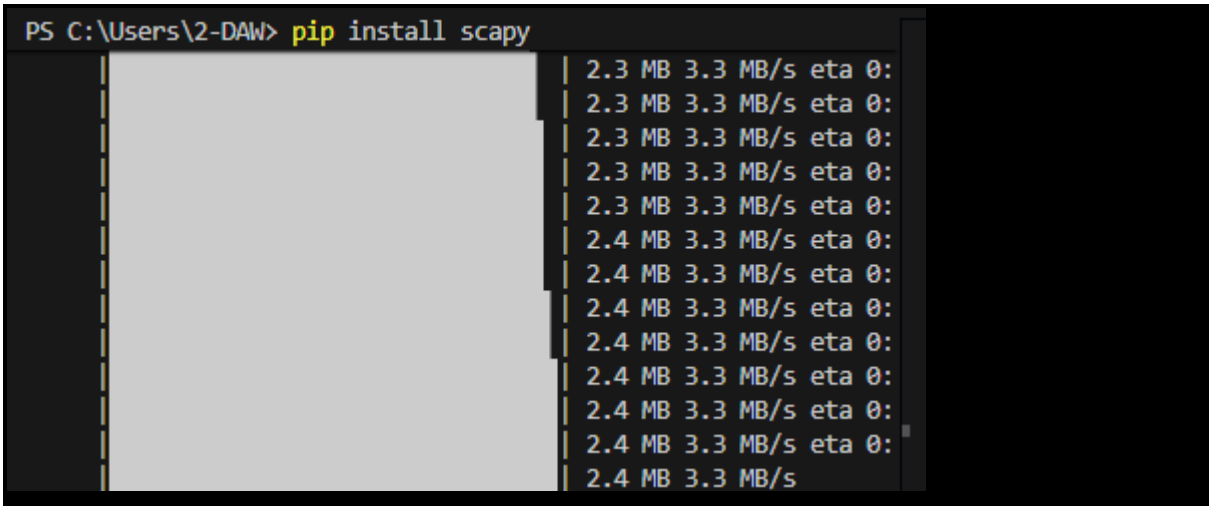
```
Installing collected packages: numpy, pandas
```

Nota legal/ética: revisa **robots.txt**, términos de uso y la ley aplicable. Limita la frecuencia de peticiones y **no** ataques servicios.

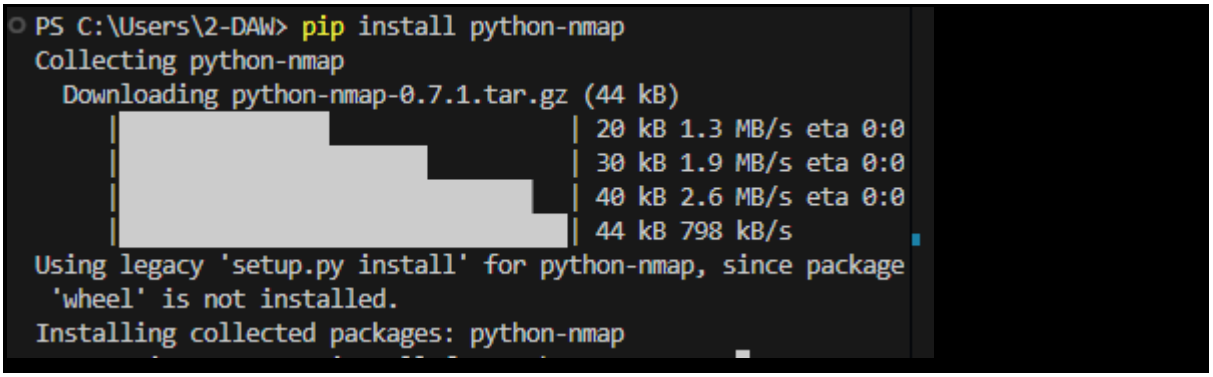
Ciberseguridad (scripting y auditoría)

- **scapy**
  Permite **construir, enviar y capturar paquetes de red** (ARP, TCP, DNS…). Útil para aprender protocolos, crear pruebas de laboratorio y automatizar diagnósticos a bajo nivel.

```
pip install scapy
```




```
pip install python-nmap
```




```
pip install impacket
```

```
PS C:\Users\2-DAW> pip install impacket
Requirement already satisfied: cffi>=1.14 in c:\users\2-daw\ap
pdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n
2kfra8p0\localcache\local-packages\python38\site-packages (fro
m cryptography<47,>=45.0.7->pyOpenSSL->impacket) (1.17.1)
Requirement already satisfied: pycparser in c:\users\2-daw\app
data\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2
kfra8p0\localcache\local-packages\python38\site-packages (from
 cffi>=1.14->cryptography<47,>=45.0.7->pyOpenSSL->impacket) (2
.23)
Using legacy 'setup.py install' for impacket, since package 'w
heel' is not installed.
Installing collected packages: MarkupSafe, Werkzeug, ldap3, Ji
nja2, itsdangerous, importlib-metadata, dnspython, blinker, py
readline3, pycryptodomex, ldapdomaindump, flask, impacket
```

pip install cryptography

```
PS C:\Users\2-DAW> pip install cryptography
Requirement already satisfied: typing-extensions>=4.13.2 in c:
\users\2-daw\appdata\local\packages\pythonsoftwarefoundation.p
ython.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\sit
e-packages (from cryptography) (4.13.2)
Requirement already satisfied: pycparser in c:\users\2-daw\app
data\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2
kfra8p0\localcache\local-packages\python38\site-packages (from
 cffi>=1.14->cryptography) (2.23)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3
```

pip install pycryptodome

```
PS C:\Users\2-DAW> pip install pycryptodome
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.7 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s eta 0:
     |                                | 1.8 MB 3.3 MB/s
```

pip install yara-python

```
PS C:\Users\2-DAW> pip install yara-python
; setup()'"'"');code = f.read().replace('"'"'\r\n'"'"', '"'"'\
n'"'"');f.close();exec(compile(code, __file__, '"'"'exec'"'"')
)' install --record 'C:\Users\2-DAW\AppData\Local\Temp\pip-rec
ord-ai7gwomt\install-record.txt' --single-version-externally-m
anaged --user --prefix= --compile --install-headers 'C:\Users\
2-DAW\AppData\Local\Packages\PythonSoftwareFoundation.Python.3
.8_qbz5n2kfra8p0\LocalCache\local-packages\Python38\Include\ya
ra-python' Check the logs for full command output.
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
```

pip install dnspython

```
PS C:\Users\2-DAW> pip install dnspython
Requirement already satisfied: dnspython in c:\users\2-daw\app
data\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2
kfra8p0\localcache\local-packages\python38\site-packages (2.6.
1)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

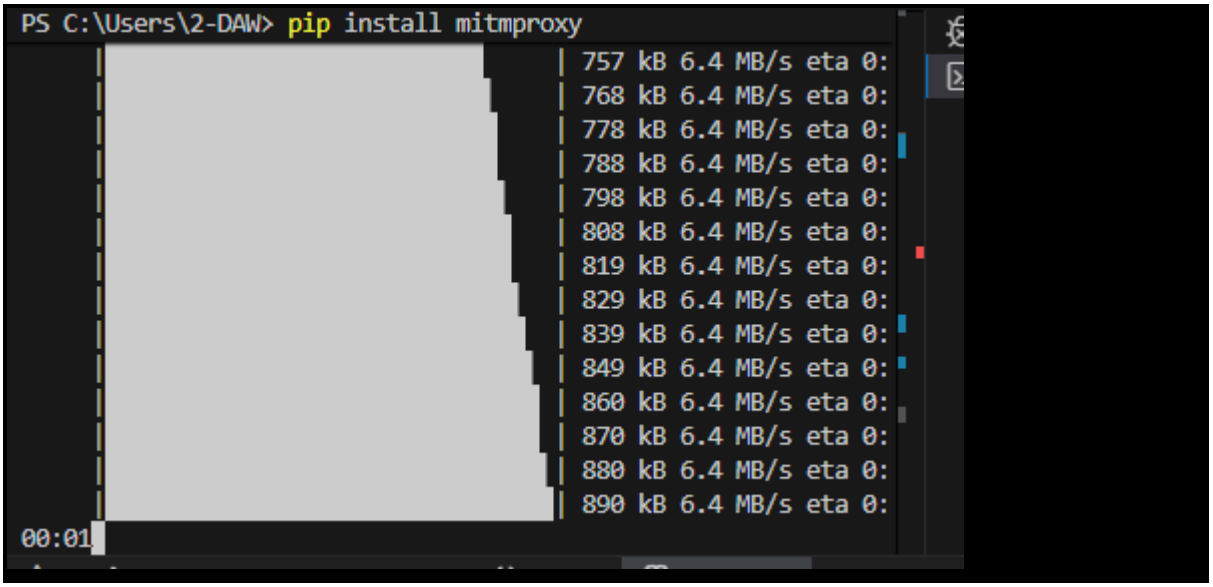
pip install shodan

```
PS C:\Users\2-DAW> pip install shodan
.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packa
ges (from requests>=2.2.1->shodan) (2025.10.5)
Requirement already satisfied: requests-file>=1.4 in c:\users\
2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3
.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packa
ges (from tldextract->shodan) (3.0.1)
Requirement already satisfied: filelock>=3.0.8 in c:\users\2-d
aw\appdata\local\packages\pythonsoftwarefoundation.python.3.8_
qbz5n2kfra8p0\localcache\local-packages\python38\site-packages
 (from tldextract->shodan) (3.16.1)
Using legacy 'setup.py install' for shodan, since package 'whe
el' is not installed.
Installing collected packages: XlsxWriter, click-plugins, shod
```

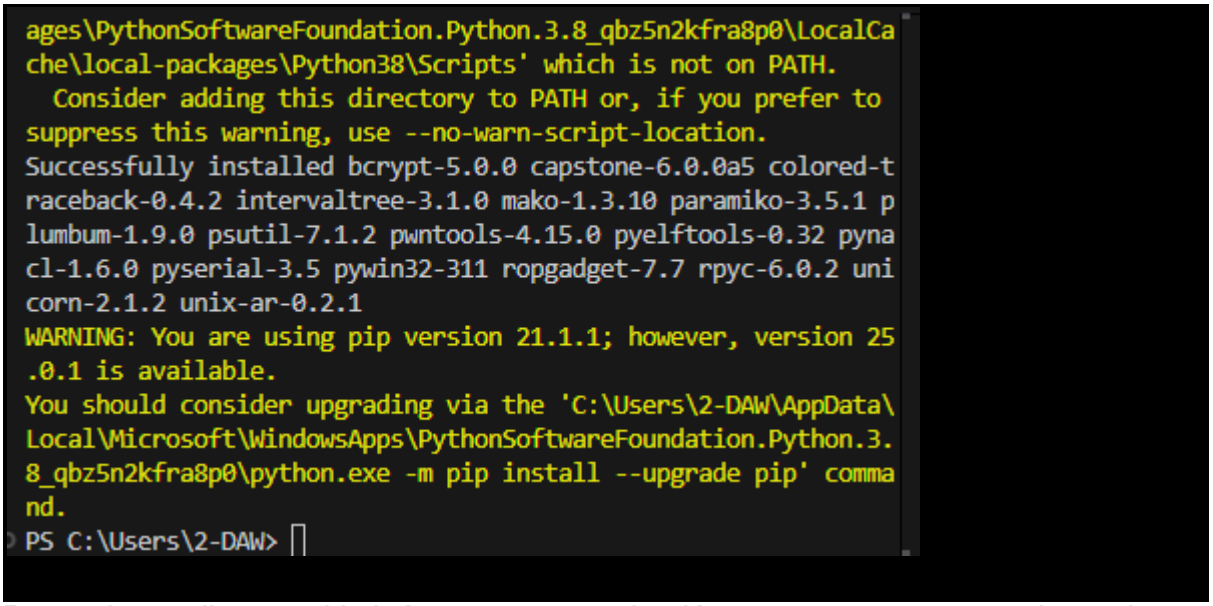
pip install censys

```
PS C:\Users\2-DAW> pip install censys
Requirement already satisfied: certifi>=2017.4.17 in c:\users\
2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3
.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packa
ges (from requests>=2.29.0->censys) (2025.10.5)
Requirement already satisfied: idna<4,>=2.5 in c:\users\2-daw\
appdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz
5n2kfra8p0\localcache\local-packages\python38\site-packages (f
rom requests>=2.29.0->censys) (3.11)
Requirement already satisfied: charset_normalizer<4,>=2 in c:\
users\2-daw\appdata\local\packages\pythonsoftwarefoundation.py
thon.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site
-packages (from requests>=2.29.0->censys) (3.4.4)
Collecting pygments<3.0.0,>=2.13.0
  Downloading pygments-2.19.2-py3-none-any.whl (1.2 MB)
     |                                        | 1.2 MB ...
```

pip install mitmproxy



pip install pwntools



Recuerda: en ciberseguridad **siempre** con autorización expresa y en entornos de pruebas.

Administración de redes / automatización IT

- **netmiko**
  Facilita abrir sesiones **SSH** a equipos de red (Cisco, HP, MikroTik, etc.) y **enviar comandos** de forma confiable. Ahorra tiempo en tareas repetitivas.

pip install netmiko

```
PS C:\Users\2-DAW> pip install netmiko
Installing collected packages: future, textfsm, scp, ntc-templ
ates, netmiko
  WARNING: The scripts futurize.exe and pasteurize.exe are ins
talled in 'C:\Users\2-DAW\AppData\Local\Packages\PythonSoftwar
eFoundation.Python.3.8_qbz5n2kfra8p0\LocalCache\local-packages
\Python38\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to
suppress this warning, use --no-warn-script-location.
  WARNING: The script textfsm.exe is installed in 'C:\Users\2-
DAW\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.8
_qbz5n2kfra8p0\LocalCache\local-packages\Python38\Scripts' whi
ch is not on PATH.
  Consider adding this directory to PATH or, if you prefer to
suppress this warning, use --no-warn-script-location.
```

pip install napalm

```
PS C:\Users\2-DAW> pip install napalm
5n2kfra8p0\localcache\local-packages\python38\site-packages (f
rom napalm) (1.17.1)
Collecting pyeapi>=1.0.2
  Downloading pyeapi-1.0.4.tar.gz (151 kB)
     |                                | 61 kB 3.8 MB/s eta 0:0
     |                                | 71 kB 4.5 MB/s eta 0:0
     |                                | 81 kB 5.1 MB/s eta 0:0
     |                                | 92 kB 5.8 MB/s eta 0:0
     |                                | 102 kB 6.4 MB/s eta 0:
     |                                | 112 kB 6.4 MB/s eta 0:
     |                                | 122 kB 6.4 MB/s eta 0:
     |                                | 133 kB 6.4 MB/s eta 0:
     |                                | 143 kB 6.4 MB/s eta 0:
     |                                | 151 kB 6.4 MB/s
```

pip install nornir

```
PS C:\Users\2-DAW> pip install nornir
Installing collected packages: mypy-extensions, importlib-meta
data, nornir
  Attempting uninstall: importlib-metadata
    Found existing installation: importlib-metadata 8.5.0
    Uninstalling importlib-metadata-8.5.0:
      Successfully uninstalled importlib-metadata-8.5.0
Successfully installed importlib-metadata-4.13.0 mypy-extensio
ns-1.1.0 nornir-3.4.1
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install pysnmp

```
PS C:\Users\2-DAW> pip install pysnmp
     |                                | 204 kB 3.3 MB/s eta 0:
     |                                | 215 kB 3.3 MB/s eta 0:
     |                                | 225 kB 3.3 MB/s eta 0:
     |                                | 235 kB 3.3 MB/s eta 0:
     |                                | 245 kB 3.3 MB/s eta 0:
     |                                | 256 kB 3.3 MB/s eta 0:
     |                                | 266 kB 3.3 MB/s eta 0:
     |                                | 276 kB 3.3 MB/s eta 0:
     |                                | 286 kB 3.3 MB/s
Requirement already satisfied: pyasn1!=0.5.0,>=0.4.8 in c:\use
rs\2-daw\appdata\local\packages\pythonsoftwarefoundation.pytho
n.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-pa
ckages (from pysnmp) (0.6.1)
Installing collected packages: pysnmp
```

pip install paramiko

```
PS C:\Users\2-DAW> pip install paramiko
Requirement already satisfied: cffi>=1.12 in c:\users\2-daw\ap
pdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n
2kfra8p0\localcache\local-packages\python38\site-packages (fro
m cryptography>=3.3->paramiko) (1.17.1)
Requirement already satisfied: pycparser in c:\users\2-daw\app
data\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2
kfra8p0\localcache\local-packages\python38\site-packages (from
 cffi>=1.12->cryptography>=3.3->paramiko) (2.23)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install netaddr

```
PS C:\Users\2-DAW> pip install netaddr
Requirement already satisfied: netaddr in c:\users\2-daw\appda
ta\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2kf
ra8p0\localcache\local-packages\python38\site-packages (1.3.0)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install speedtest-cli

```
PS C:\Users\2-DAW> pip install speedtest-cli
Installing collected packages: speedtest-cli
  WARNING: The scripts speedtest-cli.exe and speedtest.exe are
 installed in 'C:\Users\2-DAW\AppData\Local\Packages\PythonSof
twareFoundation.Python.3.8_qbz5n2kfra8p0\LocalCache\local-pack
ages\Python38\Scripts' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to
suppress this warning, use --no-warn-script-location.
Successfully installed speedtest-cli-2.1.3
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install prometheus_client

```
PS C:\Users\2-DAW> pip install prometheus_client
     |                                | 10 kB 165 kB/s eta 0:0
     |                                | 20 kB 330 kB/s eta 0:0
     |                                | 30 kB 495 kB/s eta 0:0
     |                                | 40 kB 660 kB/s eta 0:0
     |                                | 51 kB 825 kB/s eta 0:0
     |                                | 54 kB 990 kB/s
Installing collected packages: prometheus-client
Successfully installed prometheus-client-0.21.1
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

pip install requests

```
PS C:\Users\2-DAW> pip install requests
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\
2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3
.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packa
ges (from requests) (2.2.3)
Requirement already satisfied: idna<4,>=2.5 in c:\users\2-daw\
appdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz
5n2kfra8p0\localcache\local-packages\python38\site-packages (f
rom requests) (3.11)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
PS C:\Users\2-DAW>
```

Extras transversales (mejoran cualquier script)

- **loguru**
  Sistema de **logging** muy fácil de usar (niveles, rotación automática, formato bonito). Hace que registrar lo que pasa en tu script sea trivial.

```
pip install loguru
```

```
PS C:\Users\2-DAW> pip install loguru
Requirement already satisfied: colorama>=0.3.4 in c:\users\2-d
aw\appdata\local\packages\pythonsoftwarefoundation.python.3.8_
qbz5n2kfra8p0\localcache\local-packages\python38\site-packages
 (from loguru) (0.4.6)
Collecting win32-setctime>=1.0.0
  Downloading win32_setctime-1.2.0-py3-none-any.whl (4.1 kB)
Installing collected packages: win32-setctime, loguru
Successfully installed loguru-0.7.3 win32-setctime-1.2.0
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8 qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
```

```
pip install typer
```

```
PS C:\Users\2-DAW> pip install typer
Installing collected packages: shellingham, typer
  WARNING: The script typer.exe is installed in 'C:\Users\2-DA
W\AppData\Local\Packages\PythonSoftwareFoundation.Python.3.8_q
bz5n2kfra8p0\LocalCache\local-packages\Python38\Scripts' which
 is not on PATH.
  Consider adding this directory to PATH or, if you prefer to
suppress this warning, use --no-warn-script-location.
Successfully installed shellingham-1.5.4 typer-0.20.0
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
PS C:\Users\2-DAW>
```

pip install click

```
PS C:\Users\2-DAW> pip install click
Requirement already satisfied: click in c:\users\2-daw\appdata
\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2kfra
8p0\localcache\local-packages\python38\site-packages (8.1.8)
Requirement already satisfied: colorama in c:\users\2-daw\appd
ata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2k
fra8p0\localcache\local-packages\python38\site-packages (from
click) (0.4.6)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
```

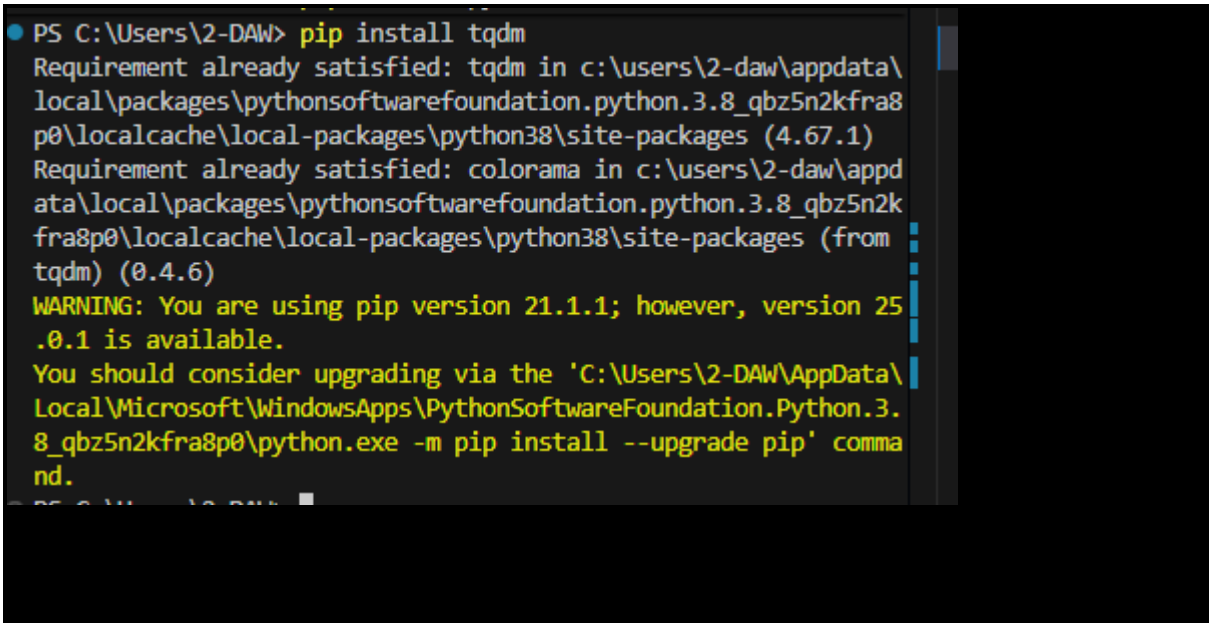
pip install rich

```
PS C:\Users\2-DAW> pip install rich
Requirement already satisfied: markdown-it-py>=2.2.0 in c:\use
rs\2-daw\appdata\local\packages\pythonsoftwarefoundation.pytho
n.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-pa
ckages (from rich) (3.0.0)
Requirement already satisfied: mdurl~=0.1 in c:\users\2-daw\ap
pdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n
2kfra8p0\localcache\local-packages\python38\site-packages (fro
m markdown-it-py>=2.2.0->rich) (0.1.2)
WARNING: You are using pip version 21.1.1; however, version 25
.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\
Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.
8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' comma
nd.
PS C:\Users\2-DAW>
```

pip install pydantic

```
PS C:\Users\2-DAW> pip install pydantic
     |                                | 368 kB 3.2 MB/s eta 0:
     |                                | 378 kB 3.2 MB/s eta 0:
     |                                | 389 kB 3.2 MB/s eta 0:
     |                                | 399 kB 3.2 MB/s eta 0:
     |                                | 409 kB 3.2 MB/s eta 0:
     |                                | 419 kB 3.2 MB/s eta 0:
     |                                | 430 kB 3.2 MB/s eta 0:
     |                                | 431 kB 3.2 MB/s
Requirement already satisfied: typing-extensions>=4.12.2 in c:
\users\2-daw\appdata\local\packages\pythonsoftwarefoundation.p
ython.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\sit
e-packages (from pydantic) (4.13.2)
Collecting annotated-types>=0.6.0
  Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
```

pip install tqdm


PS C:\Users\2-DAW> pip install tqdm
Requirement already satisfied: tqdm in c:\users\2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packages (4.67.1)
Requirement already satisfied: colorama in c:\users\2-daw\appdata\local\packages\pythonsoftwarefoundation.python.3.8_qbz5n2kfra8p0\localcache\local-packages\python38\site-packages (from tqdm) (0.4.6)
WARNING: You are using pip version 21.1.1; however, version 25.0.1 is available.
You should consider upgrading via the 'C:\Users\2-DAW\AppData\Local\Microsoft\WindowsApps\PythonSoftwareFoundation.Python.3.8_qbz5n2kfra8p0\python.exe -m pip install --upgrade pip' command.